



Department of Electrical and Electronics Engineering
Academic Year 2021 – 2022 (Odd Semester)

Degree, Semester & Branch: III Semester B.E. EEE -‘A’ Section

Course Code & Title: ME8792 Power Plant Engineering

Name of the Faculty member (s): A. S. Vigneshwar

Innovative Practice Description

- **Unit / Topic:** Unit III / Virtual Simulation of Nuclear Power Plant
- **Course Outcome:** Explain the layout, construction and working of the components inside nuclear power plants
- **Topic Learning Outcome:**
 - ✓ Explain the differences between Nuclear fission and fusion processes
 - ✓ Draw the layout of different types of Nuclear power plant
 - ✓ Illustrate the working of different types of Nuclear power plant
 - ✓ Enumerate safety measures for Nuclear power plants
- **Activity Chosen:** Virtual laboratory
- **Justification:**
 - ✓ Apply basic engineering concept in Nuclear power plant
 - ✓ Explain the working of Diesel and Gas turbine power plant
- **Time Allotted for the Activity:** 50 minutes
- **Details of the Implementation:**

Totally 31 students were actively participated for this online virtual simulation practice.

• **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
C206.3	3											

• **PO / PSO mapped:**

CO	PSO1	PSO2	PSO3
C206.3	1		

- Images / Screenshot of the practice:

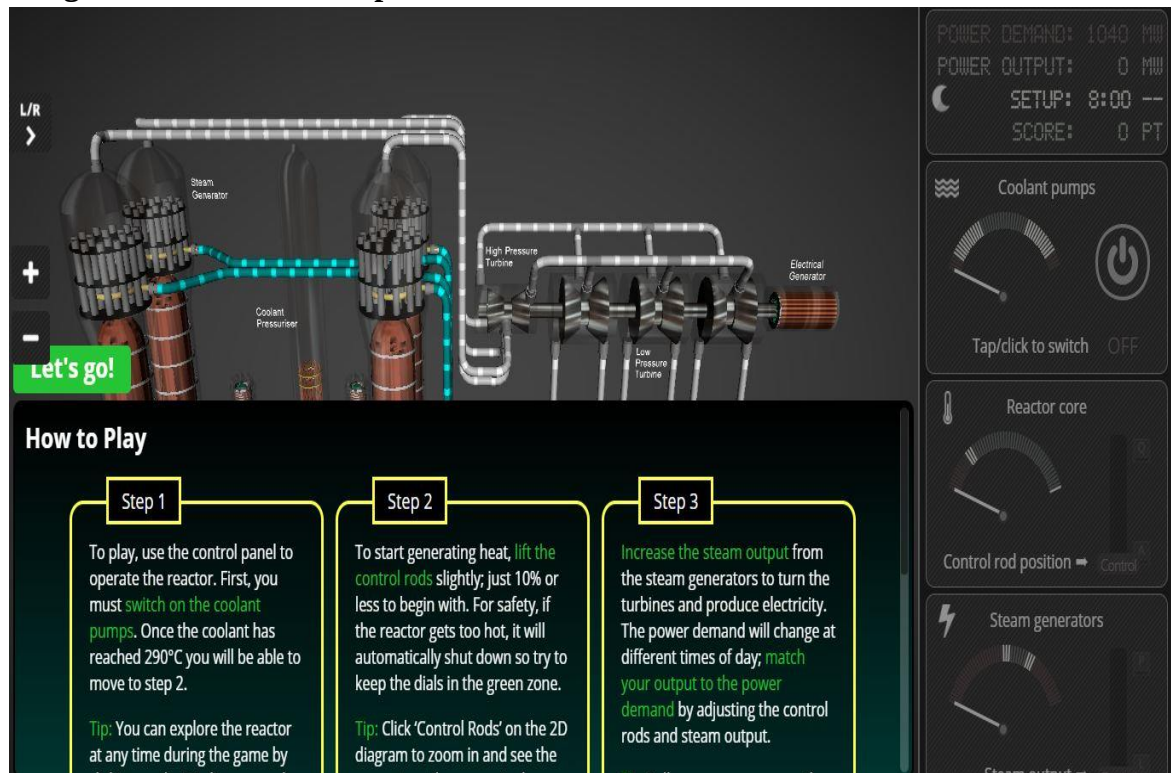


Figure1. “Illustrate the typical layout of nuclear power plant”

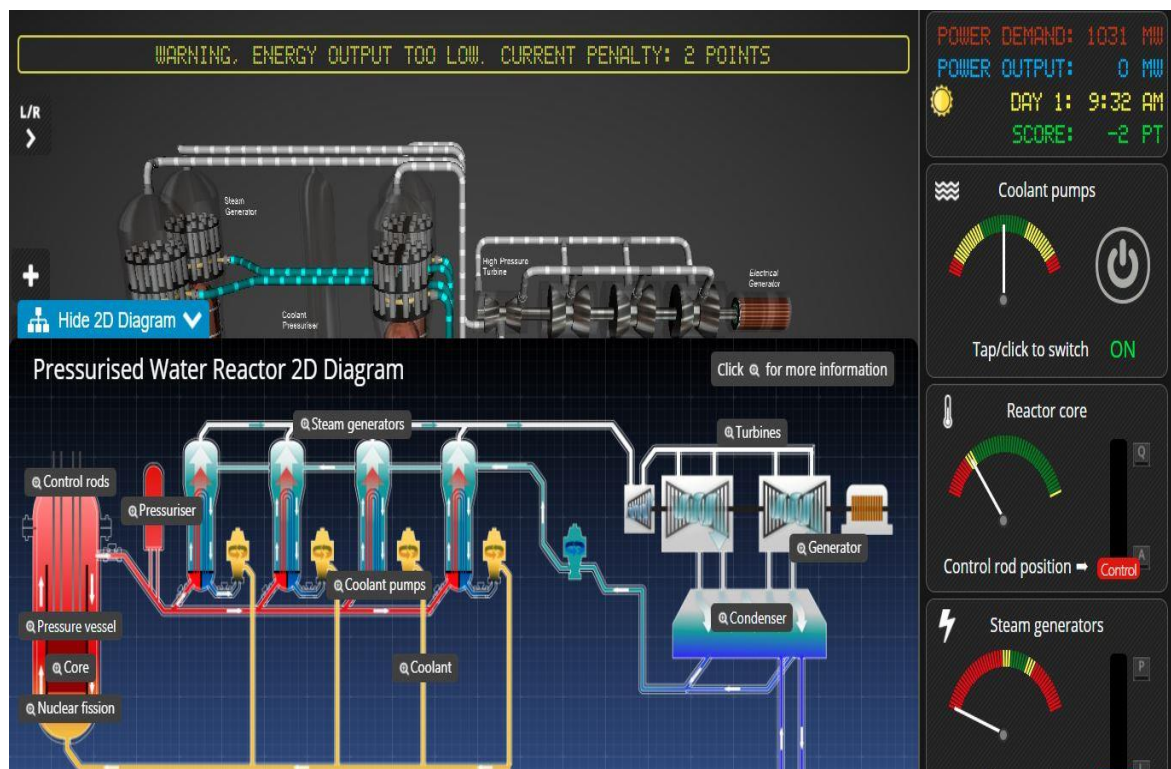


Figure2. “Illustrate the working of PWR”

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- The overall feedback from the students were really good and students felt that it is very easy understand the real time operation of nuclear power plant

- ❖ ***Benefit of the practice:***

- After completing this hands on session in virtual laboratory, the students were able to understand the operation of nuclear power plant very clearly. This virtual simulation enables the students to better understanding of their real time operation of nuclear power plant by adjusting the control rods, fuel inputs and coolants used.

- ❖ ***Challenges faced in implementation:***

- The warning and annunciation of fuel input provision can be included in simulation, for better clarity

References:

- ❖ <http://lotus.itservices.manchester.ac.uk:8080/>



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Course Code & Title: ME8792 Power Plant Engineering

Name of the Faculty member (s): A. S. Vigneshwar

Innovative Practice Description

- **Unit / Topic:** Unit IV / Virtual Simulation of Renewable Energy Power Plant
- **Course Outcome:** Explain the layout, construction and working of the components inside Renewable energy power plants.
- **Topic Learning Outcome:**
 - ✓ Classify hydroelectric power plant
 - ✓ Explain the working of the different renewable energy systems
 - ✓ Draw the layout of different renewable energy systems
 - ✓ Explain the working of different types of fuel cell and its applications
- **Activity Chosen:** Virtual laboratory
- **Justification:**
 - ✓ Apply basic engineering concept in Renewable energy systems
 - ✓ Explain the advantages of renewable energy system based on environmental issue
 - ✓ Apply the fundamental of renewable energy sources in project
 - ✓ Explain the working of different renewable energy systems.
- **Time Allotted for the Activity:** 50 minutes
- **Details of the Implementation:**

Totally 31 students were actively participated for this online virtual simulation practice.

• **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
C206.4	3						3					1

• **PO / PSO mapped:**

CO	PSO1	PSO2	PSO3
C206.4	1		

- **Images / Screenshot of the practice:**

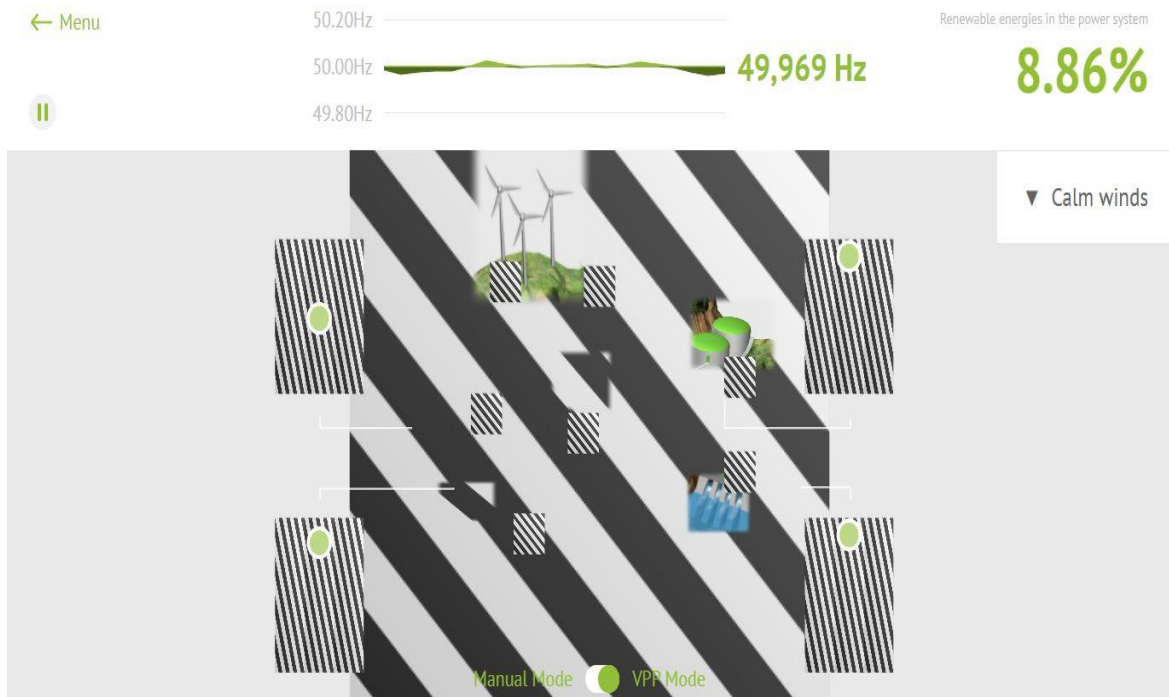


Figure1. “Illustrate the operation of renewable energy power plant”

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

The overall feedback from the students were really good and students felt that it is very easy understand the real time operation of renewable energy power plant

- ❖ ***Benefit of the practice:***

- ✓ After completing this hands on session in virtual laboratory, the students were able to understand the operation of renewable power plant very clearly.
- ✓ This virtual simulation enables the students to better understanding of their real time operation of renewable energy power plants such as Solar and wind by adjusting the frequency of the system to grid.
- ✓ This virtual simulation also helps the students to know the stability of the system.

- ❖ ***Challenges faced in implementation:***

The output of simulation part can be better.

References:

- ❖ <https://www.next-kraftwerke.com/virtual-power-plant-vpp-simulation/?lang=en>